Matrix functions and stability

A uniform disk bound for wave-kernel Padé approximants

Difficulty: hard **Importance:** interesting to specialist

Status: Partially resolved

Rating rationale: Hard because this is a focused all-orders Padé inequality within established approximation theory; its immediate impact is on specialist wave-kernel error analysis.

Problem statement

Let

$$f(z) = \sum_{j=0}^{\infty} \frac{z^j}{(2j)!} = \cosh \sqrt{z},$$

where the series defines the entire function without a square-root branch choice. For each integer $m \geq 1$, let $r_m = P_m/Q_m$ be its diagonal Padé approximant at zero: $\deg P_m, \deg Q_m \leq m$, $Q_m(0) = 1$, and $Q_m(z)f(z) - P_m(z) = O(z^{2m+1})$. Is it true that the reduced rational function r_m has no pole in $\{z \in \mathbb{C} : |z| \leq 3\}$ and

$$|1 - r_m(z)| \leq 2 \qquad (|z| \leq 3)$$

for every m?

Reference and status evidence

Nadukandi and Higham, Computing the Wave-Kernel Matrix Functions, SIAM J. Scientific Computing 40(6) (2018), DOI, §4.2, Conjecture 4.6, manuscript p. 12. Lemma 4.5 establishes the finite range $m \leq 20$; §4.3 explains its role in backward-error analysis. Searches for the paper title with "conjecture" and for "Conjecture 4.6" with "cosh" and "Padé" located no general proof or counterexample. The status evidence is therefore weaker than a recent paper explicitly reaffirming the conjecture.

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Rechecked Lemma 4.5 and Conjecture 4.6: the displayed assertion is proved for $1 \leq m \leq 20$, while arbitrary m remains conjectural. Paper-title and conjecture-number searches found no general resolution. The open portion still rests on historical primary evidence.